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Exercise: Day 1

In this problem, you will consider a simple model of the conformational statistics of a polymer chain. This idealized model consists of N non-interacting segments connected end-to-end. Each segment, i , has a fixed length, ℓ , and an orientation given by the unit vector, $\hat{b}_i$ that can take one of 6 values, $\pm\hat{x}$, $\pm\hat{y}$, or $\pm\hat{z}$, where $\hat{x}$, $\hat{y}$, and $\hat{z}$ denote the orthogonal cartesian directions. In essence, chain configurations trace out a random walk on a cubic lattice. In the idealized model, assume that the orientations of segments are statistically independent (*i.e.*, $\langle \hat{b}_i \cdot \hat{b}_j \rangle = 0$ for $i \neq j$) and that each direction is equally likely (*i.e.*, $\langle \hat{b}_i \rangle = 0$). Segments do not interact and can thus occupy the same lattice site.

1. Show that the entropy of this N -unit ideal chain has the form $S = Nk_B \ln a$, and determine a . How does a depend on the dimensionality?
2. Calculate the mean squared end-to-end distance of the chain, $\langle R^2 \rangle$, where $R = |\mathbf{R}|$, and
$$\mathbf{R} = \sum_{i=1}^N \ell \hat{b}_i.$$
 Your result should reveal that the typical distance between the ends of the chain scales with chain length as, $\sqrt{\langle R^2 \rangle} \sim N^\nu$. Identify the value of the exponent, ν and its dependence on dimensionality.
3. For a real polymer in a good solvent, experiments indicate that $\nu = 3/5$. Compare this value to the result of question 2 and comment on the discrepancy (if there is one).
4. Now imagine that adjacent segments are subject to an external potential energy $u = -\gamma \hat{b}_i \cdot \hat{a}$. Compute the mean segment orientation, $\langle \hat{b}_i \rangle$, for a chain subject to this potential energy. In addition, compute $\langle R^2 \rangle$ and plot its dependence on the parameter, γ . For this calculation, assume that $\hat{a}$ is aligned with one of the Cartesian directions. Would you expect the result to change if $\hat{a}$ wasn't aligned with the underlying lattice?